Handout

Physics Grade 8 - Thermal Expansion

Thermal expansion is the tendency of matter to change its dimensions due to temperature changes. Almost all materials expand when heated and contract when cooled. This behavior plays a crucial role in engineering, construction, and everyday life. Understanding thermal expansion helps prevent structural damage and explains how many devices work.

Fig 1. Rail buckling, driven by thermal expansion in lengthy continuous track sections, caused 190 train derailments across the US from 1998–2002.

Why Does Thermal Expansion Happen?

The main reason thermal expansion occurs is the motion of particles within matter. All substances are made up of tiny particles that are constantly moving. When a material is heated, these particles gain kinetic energy and vibrate more rapidly. As they vibrate faster, the distance between the particles increases, causing the material to expand. When the material is cooled, the particles lose energy, move more slowly, and come closer together, resulting in contraction. Keep in mind that the particles themselves do not grow in size, only the space between them changes.

Thermal expansion is the increase in the size of a substance when its temperature rises. This increase can occur in length, area and volume. When temperature decreases, the substance undergoes thermal contraction, which is the decrease in size.

Thermal Expansion In Solid, Liquid and Gas

Thermal expansion behaves differently in solids, liquids, and gases. Solids expand the least because their particles are tightly packed and strongly bonded. Liquids expand more than solids, which is why

they are often used in thermometers. Gases expand the most because their particles are far apart and move freely, making their volume very sensitive to temperature changes.

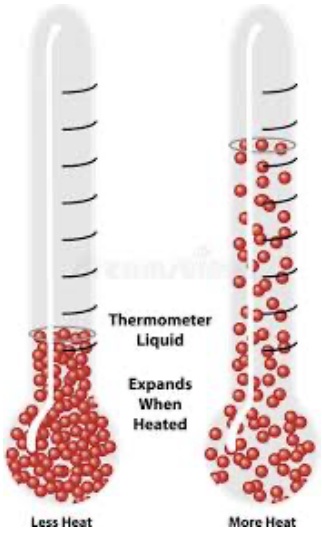

Fig 2. When temperature increases, the liquid inside a thermometer expands and rises, while cooling causes it to contract and fall, demonstrating liquid thermal expansion.

Thermal Expansion In Solid

Thermal expansion can occur in different forms depending on how the size of the object changes. Linear expansion refers to a change in length and is most noticeable in long solid objects such as metal rods or railway tracks.

The amount a material expands depends on its **coefficient of thermal expansion**, which is unique to each material. Materials with a larger coefficient expand more for the same temperature increase. Metals generally have higher expansion coefficients than materials like glass or ceramics. Because of this.

The amount of linear expansion depends on three main factors: the **original length of the object**, the **change in temperature**, and the **type of material**. Longer objects expand more than shorter ones under the same temperature change. Similarly, a greater temperature increase causes more expansion. Different materials expand by different amounts depending on their **coefficient of linear expansion**, which indicates how much a material expands per degree of temperature change.

The linear expansion is represented by this formula:

$$\Delta L = \alpha L_o \Delta T$$

where ΔL is the change in length, L_0 is the original length, α is the coefficient of linear expansion, and ΔT is the change in temperature. This formula shows that linear expansion is directly proportional to the original length and the temperature change.

Thermal Expansion In Gas

Thermal expansion in gas causes particles to collide more frequently and more forcefully with the walls of the container. As a result, the gas either expands in volume or increases in pressure, depending on whether the container can change size.

When a gas is kept in a **rigid (fixed-volume) container**, it cannot expand. In this case, heating the gas causes the **pressure to increase** instead. This relationship between pressure and temperature is described by the formula

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

which is known as **Gay-Lussac's law**. It states that for a fixed amount of gas at constant volume, the pressure of a gas is directly proportional to its absolute temperature.

In the formula, P_1 and P_2 represent the initial and final pressures of the gas, while T_1 and T_2 represent the initial and final temperatures measured in **kelvin (K)**. Using kelvin is important because this temperature scale starts at absolute zero, where particle motion theoretically stops. As temperature increases in kelvin, the pressure increases in the same ratio.

Thermal Expansion In Liquid

Water is a special case in thermal expansion. Unlike most substances, water expands when cooled from 4°C to 0°C and contracts when heated from 0°C to 4°C. It reaches its maximum density at 4°C. Above this temperature, it expands normally. The phenomenon is called **anomalous expansion of water.** This unusual behavior allows ice to float on water and helps aquatic life survive in cold environments.